

C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	System Generated	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	User Input	The user types this value directly at the keyboard when prompted by the system.
Default Value	Default Value	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	Calculated	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	From List	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: <u>book</u>		
Property Name	Data Type	Source Tag
<u>bookId</u>	<u>int</u>	<u>system generated</u>
<u>title</u>	<u>string</u>	<u>user input</u>
<u>author</u>	<u>string</u>	<u>user input</u>
<u>genre</u>	<u>string</u>	<u>user input</u>
<u>totalCopies</u>	<u>int</u>	<u>user input</u>
<u>totalBorrowCount</u>	<u>int</u>	<u>calculated</u>
<u>availableCopies</u>	<u>int</u>	<u>calculated</u>

Entity Name: <u>Member</u>		
Property Name	Data Type	Source Tag
<u>memberId</u>	<u>int</u>	<u>System Generated</u>
<u>memberName</u>	<u>string</u>	<u>user input</u>
<u>memberPhone</u>	<u>string</u>	<u>user input</u>
<u>memberEmail</u>	<u>string</u>	<u>user input</u>
<u>registrationDate</u>	<u>string</u>	<u>System Generated</u>
<u>isActive</u>	<u>bool</u>	<u>Default Value</u>
<u></u>	<u></u>	<u></u>

Entity Name: <u>Reservation</u>		
Property Name	Data Type	Source Tag
<u>reservationId</u>	<u>int</u>	<u>System Generated</u>
<u>memberId</u>	<u>int</u>	<u>from list</u>
<u>bookId</u>	<u>int</u>	<u>from list</u>
<u>reservationDate</u>	<u>string</u>	<u>System Generated</u>
<u>status</u>	<u>string</u>	<u>Default Value</u>
<u></u>	<u></u>	<u></u>
<u></u>	<u></u>	<u></u>

Entity Name: <u>Borrow</u>		
Property Name	Data Type	Source Tag
<u>BorrowId</u>	<u>int</u>	<u>System Generated</u>
<u>memberId</u>	<u>int</u>	<u>From List</u>
<u>bookId</u>	<u>int</u>	<u>From List</u>
<u>borrowDate</u>	<u>string</u>	<u>System Generated</u>
<u>dueDate</u>	<u>string</u>	<u>Calculated</u>
<u>daysOverdue</u>	<u>int</u>	<u>Calculated</u>
<u>fineAmount</u>	<u>decimal</u>	<u>Calculated</u>

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

1. When the member is registered, the system creates a member record with memberId, memberName, memberPhone, memberEmail, registrationDate, and isActive.

2. Each Book has bookId, title, author, genre, total copies, totalBorrowCount, and availableCopies. The available copies shows how many copies can be borrowed.

3. When a Member wants to borrow a Book the user select the memberId and the bookId from the lists. The system creates a Borrow record. The borrowDate is generated by the system, and the dueDate is calculated as 14 days after the borrowDate

4. the book available copies decreases by 1. Also the total borrow count for that book increases by 1 because the book has been borrowed

5. If member returns the book late, the system calculates days overdue

6. If book has no available copies, the member can reserve it. then the system creates reservation record using memberId and bookId

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	RegisterMember	add new member to the library system
2	AddBook	add new book with title, author, genre, and number of copies
3	ViewAllBooks	displays all books with their details
4	ViewAllMembers	displays all registered library members
5	BorrowBook	creates a borrow record when member borrows available book
6	ReturnBook	updates the borrow record when the member returns book
7	CalculateDueDate	calculates due date as 14 days after borrow date
8	CalculateFine	calculates the fine amount if the book is returned late
9		
10		

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest		
Property Name	Data Type	Source Tag (fill in)
<code>guestId</code>	int	System Generated
<code>guestName</code>	string	User Input
<code>guestPhone</code>	string	User Input
<code>guestEmail</code>	string	User Input
<code>guestNationality</code>	string	User Input
<code>registrationDate</code>	string	System Generated

Room		
Property Name	Data Type	Source Tag (fill in)
roomId	int	_____ System Generated
roomNumber	string	_____ User Input
roomType	string	_____ From List
floorNumber	int	_____ User Input
nightlyPrice	decimal	_____ Calculated
isAvailable	bool	_____ Default Value

Booking		
Property Name	Data Type	Source Tag (fill in)
bookingId	int	_____ System Generated
guestId	int	_____ From List
roomId	int	_____ From List
checkInDate	string	_____ User Input
checkOutDate	string	_____ User Input
numberOfNights	int	_____ User Input
totalPrice	decimal	_____ Calculated
status	string	_____ Default Value

CheckInRecord		
Property Name	Data Type	Source Tag (fill in)
recordId	int	System Generated
bookingId	int	From List
guestId	int	From List
roomId	int	From List
actualCheckIn	string	System Generated
actualCheckOut	string	System Generated
finalAmount	decimal	Calculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

- 1.guest registers in the hotel system. When registration happens, the system creates a Guest object and generates the guestId and registrationDate
- 2.room created with room number, floor number, and room type. user select the room Type from a list such as single, double, or suite. nightly Price calculated from the selected room type, and isAvailable starts as true.
- 3.system calculates number of nights from checkIn date and checkout date, then calculates total price using number of nights
- 4.after booking, the room isAvailable change to false

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	RegisterGuest	adds a new guest to the hotel system
2	ViewAllGuests	displays all registered guests
3	AddRoom	add new rooms with type, floor number, and price
4	ViewRooms	Displays all rooms
5	ViewAvailableRooms	displays only rooms that are available
6	CalculateNumberOfNights	calculates the number of nights between check-in and check-out dates
7	CalculateTotalPrice	calculates total price using number of nights and room nightly price

#	Service Name	What it does
8	ViewActiveBookings	displays booking that are confirmed.
9	HotelBookingReport	Display booking details